49. The Isomorphism Extension Theorem

1. (See the answer in the text.)

2. Extensions are τ_1 given by

$$\tau_1(\sqrt{2}) = \sqrt{2}, \quad \tau_1(\sqrt{3}) = -\sqrt{3}, \quad \tau_1(\sqrt{5}) = \sqrt{5},$$

and τ_2 given by

$$\tau_2(\sqrt{2}) = \sqrt{2}, \quad \tau_2(\sqrt{3}) = \sqrt{3}, \quad \tau_2(\sqrt{5}) = -\sqrt{5}.$$

3. (See the answer in the text.)

4. The extensions are the identity map of $\mathbb{Q}(\sqrt[3]{2})$ onto itself, and τ_1 given by $\tau_1(\alpha_1) = \alpha_2$, that is, the map $\psi_{\alpha_1, \alpha_2}$, and $\tau_2 = \psi_{\alpha_1, \alpha_3}$.

5. (See the answer in the text.)

6. The extensions are

$$\tau_1 \text{ given by } \tau_1(i) = i, \quad \tau_1(\sqrt{3}) = -\sqrt{3}, \quad \tau_1(\alpha_1) = \alpha_1,$$

$$\tau_2 \text{ given by } \tau_2(i) = i, \quad \tau_2(\sqrt{3}) = -\sqrt{3}, \quad \tau_2(\alpha_1) = \alpha_2,$$

$$\tau_3 \text{ given by } \tau_3(i) = i, \quad \tau_3(\sqrt{3}) = -\sqrt{3}, \quad \tau_3(\alpha_1) = \alpha_3,$$

$$\tau_4 \text{ given by } \tau_4(i) = -i, \quad \tau_4(\sqrt{3}) = -\sqrt{3}, \quad \tau_4(\alpha_1) = \alpha_1,$$

$$\tau_5 \text{ given by } \tau_5(i) = -i, \quad \tau_5(\sqrt{3}) = -\sqrt{3}, \quad \tau_5(\alpha_1) = \alpha_2,$$

$$\tau_6 \text{ given by } \tau_6(i) = -i, \quad \tau_6(\sqrt{3}) = -\sqrt{3}, \quad \tau_6(\alpha_1) = \alpha_3.$$

7. (See the answer in the text.)

8. F T F T F T T T T F

9. Now $\sigma : K \to K$ is an isomorphism, so $\sigma^{-1} : \sigma[K] \to K$ is an isomorphism. Because K is algebraically closed and is algebraic over $\sigma[K]$, Theorem 49.3 shows that σ^{-1} has an extension to an isomorphism τ mapping K onto a subfield of K. But σ^{-1} is already onto K, and because τ must be a one-to-one map, we see that it cannot be defined on any elements of K not already in $\sigma[K]$. Thus $\sigma[K] = K$, so σ is an automorphism of K.

10. Let E be an algebraic extension of F and let τ be an isomorphism of E onto a subfield of $\overline{F}$ that leaves F fixed. Because E is an algebraic extension of F, the field $\overline{F}$ is an algebraic extension of E and is an algebraic closure of E. By Theorem 49.3, τ can be extended to an isomorphism σ of $\overline{F}$ onto a subfield of $\overline{F}$. By Exercise 9, such an isomorphism σ is an automorphism of $\overline{F}$.

11. By Theorem 49.3, the identity map of F onto F has an extension to an isomorphism τ mapping E onto a subfield of $\overline{F}$. By Theorem 49.3, τ can be extended to an isomorphism σ mapping $\overline{E}$ onto a subfield of $\overline{F}$. Then σ^{-1} is an isomorphism mapping $\sigma[\overline{E}\,]$ onto $\overline{F}$. By Theorem 49.3, σ^{-1} can be extended to an isomorphism of $\overline{F}$ onto a subfield of $\overline{E}$. Because σ^{-1} is already onto $\overline{E}$ and its extension must be one to one, we see that the domain of σ^{-1} must already be $\overline{F}$. Thus $\sigma[\overline{E}\,] = \overline{F}$ and σ is an isomorphism of $\overline{E}$ onto $\overline{F}$.